Deutsch-Jozsa Algorithm Summary

Two types of functions come to mind when talking about this algorithm:

- **Constant function:** A function that always return the same bit as the input bit (returns 0 or 1 for all input)
- **Balanced function:** A function that always return a different bit as the input bit (returns 0 for half the input and 1 for the other half)

Classical Computers need to query more than once to check if the function is balanced or constant. We input both 0 and 1 so we can check if f(0) = f(1) or not. *Quantum Computers* on the other hand only need one query of the function to determine if it is constant or balanced.

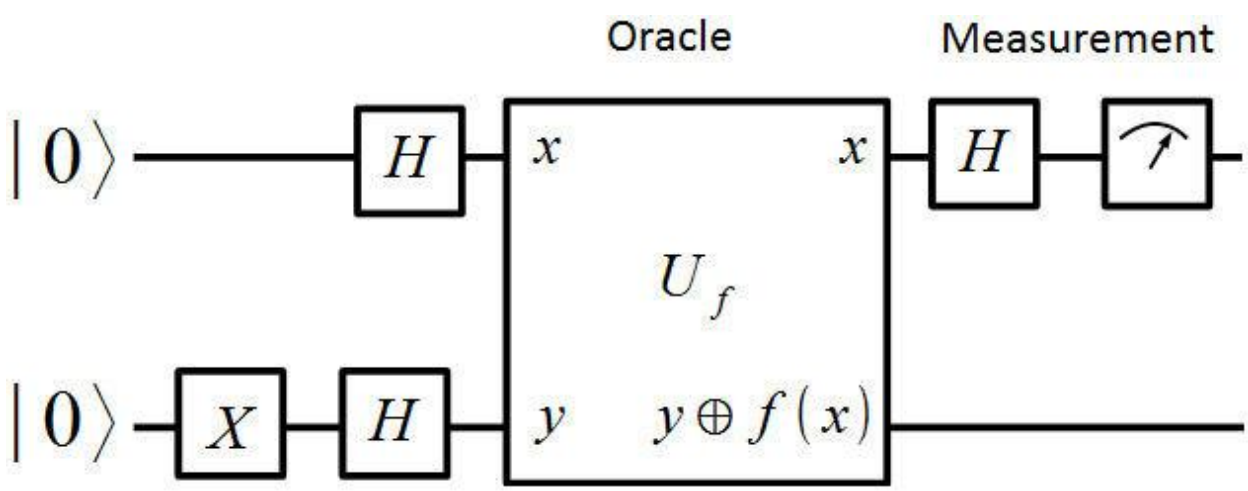
The Deutsch and Deutsch-Jozsa's algorithms are the quantum algorithms that help us query to determine if the function is constant or balanced, what is their difference though? And how do they work?

Deutsch's Algorithm

This algorithm helps us determine whether a function f:{0,1}→{0,1} is **constant** (returns the same output for both inputs) or **balanced** (returns different outputs for each input). It can only be applied to a single bit.

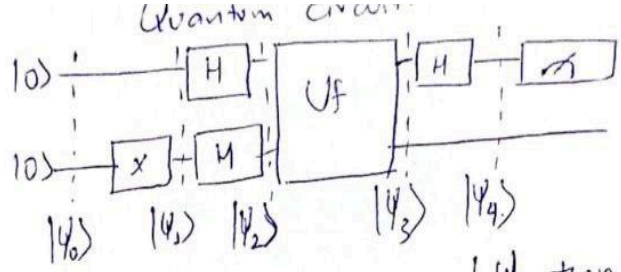
- Classical Computers would need to check twice for this case.

The Quantum Circuit:



Proof:

Quantum circuit



$|\psi_0\rangle = |00\rangle$

$|\psi_1\rangle = |01\rangle$

$|\psi_2\rangle = |+\rangle|-\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|-\rangle$

$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle|-\rangle)$

$|\psi_3\rangle = U_f\,|\psi_2\rangle$

$= \frac{1}{\sqrt{2}}(U_f|0\rangle|-\rangle + U_f|1\rangle|-\rangle)$

⊗ $U_f\,|x\rangle|-\rangle$ --- Let's determine this first

$\rightarrow U_f.|x\rangle\left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$

$\rightarrow |x\rangle\left(\frac{1}{\sqrt{2}}(U_f|0\rangle - U_f|1\rangle)\right)$

$\rightarrow |x\rangle\left(\frac{1}{\sqrt{2}}(|0 \otimes f(x)\rangle - |1 \otimes f(x)\rangle)\right)$

$\rightarrow |x\rangle\left(\frac{1}{\sqrt{2}}(|f(x)\rangle - |\overline{f(x)}\rangle)\right)$

$\rightarrow \frac{1}{\sqrt{2}}(|x\rangle|f(x)\rangle - |x\rangle|\overline{f(x)}\rangle)$

$\rightarrow \begin{cases} \frac{|x\rangle}{\sqrt{2}}(|0\rangle - |1\rangle) & f(x) = 0 \\ \frac{|x\rangle}{2}(|1\rangle - |0\rangle) & f(x) = 1 \end{cases}$

$\rightarrow \begin{cases} |x\rangle|-\rangle & , f(x) = 0 \\ -|x\rangle|-\rangle & , f(x) = 1 \end{cases} = -1^{f(x)}|x\rangle|f(x)\rangle$

ψ_3 then becomes

$\frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0\rangle|-\rangle + (-1)^{f(1)}|1\rangle|-\rangle\right)$

if $f(0) = f(1)$

$\psi_3 = |+\rangle$

if $f(0) \neq f(1)$

$\psi_3 = |-\rangle$

ψ_4, we have two cases

if $\psi_3 = |+\rangle$

$\psi_4 = |0\rangle$

if $\psi_3 = |-\rangle$

$\psi_4 = |1\rangle$

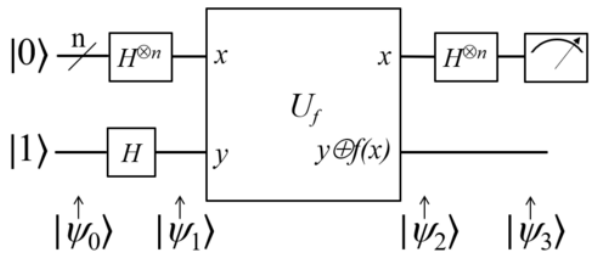
⊗ Therefore if the final measurement is $|0\rangle$ it is constant, else it is balanced

Deutsch-Jozsa Algorithm

Similar to the previous algorithm it is used to determine whether a function $f:\{0,1\}^n \rightarrow \{0,1\}$ is **constant** (same output for all inputs) or **balanced** (outputs 0 for half and 1 for half), assuming it's guaranteed to be one or the other. The only difference is that it can be applied to any number of bits.

- Classical Computers would need to check $2^{(n-1)} + 1$ times for any number of bits.

The Quantum Circuit and proof:



$$|\psi_0\rangle = |0\rangle^{\otimes n}|-\rangle$$

$$|\psi_1\rangle = H^{\otimes n}|0\rangle^{\otimes n}|-\rangle$$

$$H^{\otimes n}|0\rangle^{\otimes n} = H|0\rangle H|0\rangle \ldots H|0\rangle$$

$$\boxed{H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle}$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle|-\rangle$$

$$\psi_2 = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} U_f|x\rangle|-\rangle$$

$$\psi_2 = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)}|x\rangle$$

$$\psi_3 = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} H^{\otimes n}|x\rangle$$

Let's find $H^{\otimes n}|x\rangle$

$$H|x_i\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + (-1)^{x_i}|1\rangle\right)$$

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z\in\{0,1\}^n} (-1)^{x\cdot z}|z\rangle$$

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{z\in\{0,1\}^n} (-1)^{x\cdot z}|z\rangle$$

⇒ If f is constant

- $\frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} = 1$, if $f(x) = 0$
- $\frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} = -1$, if $f(x) = 1$

∴ If f is constant the final output would be ± 1

⇒ If f is balanced.

The final output would be 0 because equal amount of 1's and -1's will be added to give us 0